“Doctor, look into my files
I've been pipin' code, but there's no reads compiled.
Doctor, the problem's in my test,
My script runs cold as ice - just like my self-confidence”

~ Cache the elephant ~

1. Assume you have 8 .fastq’s:

   Sample_1_R1.fastq
   Sample_1_R2.fastq
   Sample_2_R1.fastq
   Sample_2_R2.fastq
   Sample_3_R1.fastq
   Sample_3_R2.fastq
   Sample_4_R1.fastq
   Sample_4_R2.fastq

   Align them to the reference genome (hisat2). Assume the names of the new files produced have the same name as the fastq’s, but the extension .sam.

2. Make the sams bams, sort and index them (samtools). Assume the names of the new files produced have the same name as the fastq’s, but the extension _sorted.bam.

3. Remove the sams (we only need the bam files).

4. Infer the library strandness (infer_experiment.py). Assume they are stranded on the negative (reversed) strand.

5. Find the counts (htseq-count). Assume the names of the new files produced, have the same name as the fastq’s, but the extension _counts.txt. Make sure to add this flag --additional-attr=gene_name

Make one bash script for all those steps (do not lose your mind with the various flags, if it gets really complicated with the flags, watch Lilo and stitch (ifelse)).

Absolute paths:

- **Reference genome (this suffices, you do not need the extension part):** /stanley/kubrick/Genomes/mm10/Hisat2Index/mm10/genome

- **Bed file (Its like the GTF file, and we use it in infer_experiment.py):** /michel/gondry /mm10_RefSeq.bed

- **GTF file:** /wes/anderson/mm10/Mus_musculus.GRCm38.101.ccds.modified.gtf

FINALLY assume the output directory is your current directory (./)

!!!! Do not forget the sanity checks with echo !!!!
!!!! Save the paths to variables !!!!
!!!! Paired end ???? !!!!